

Advanced Persistent Threats: Attack Stages, Examples, and Mitigation

What Are Advanced Persistent Threats? (APTs)

An Advanced Persistent Threat (APT) refers to a complex and covert cyber-attack executed by highly skilled threat actors, usually targeting high-profile organizations. APTs are often backed by nation-states or criminal organizations and can remain undetected within the victim's network for extended periods, ranging from months to years.

The primary objective of an APT attack is to infiltrate a network without authorization and maintain persistent access, while collecting valuable data or compromising vital systems. Organizations and governments face significant risks from these attacks, as they can result in considerable losses, including financial damage, tarnished reputations, and stolen sensitive data.

This is part of a series of articles about [cybersecurity attacks](#)

What are the Unique Characteristics of Advanced Persistent Threats?

APTs are distinguished from other cyber threats by their unique characteristics, which include:

- **High level of sophistication:** APTs are characterized by their use of advanced tools, tactics, and techniques that are designed to evade detection and bypass security measures. This often involves custom malware, zero-day exploits, and advanced social engineering tactics.
- **Targeted attacks:** APTs are usually aimed at specific organizations, industries, or countries. The attackers carefully select their targets based on strategic objectives, such as stealing intellectual property, compromising critical infrastructure, or gaining a competitive advantage.

- **State sponsorship or well-funded organizations:** APTs are often attributed to nation-states or well-funded criminal organizations that have the resources, expertise, and motivation to carry out these advanced attacks.
- **Long-term approach:** APTs are typically designed for long-term operations, with the attackers focusing on maintaining a persistent presence within the target network. This allows them to gather intelligence, exfiltrate data, or cause damage over an extended period.

Multi-stage and multi-vector: APTs usually involve multi-stage attacks that progress through various phases, such as reconnaissance, weaponization, delivery, exploitation, installation, command and control, and execution. Attackers may use multiple vectors to achieve their objectives, including spear phishing, supply chain compromise, and watering hole attacks.

Stages of an APT Attack

Advanced Persistent Threat (APT) attacks involve a complex process where attackers use multiple tactics and techniques to infiltrate a target's network, maintain persistence, and steal data:

1. **Reconnaissance:** In this first stage, attackers collect information about the target organization by investigating its infrastructure, employees, partners, or customers. They may employ open-source intelligence (OSINT), social engineering tactics, or exploit known vulnerabilities in publicly accessible systems.
2. **Infiltration:** After gathering enough information about the target environment and its vulnerabilities, attackers deploy customized malware or use other attack vectors like spear-phishing emails to gain unauthorized access to the system.
3. **Establishing foothold:** Upon entering the network, usually by exploiting vulnerabilities or using compromised credentials from phishing attacks, attackers create a foothold by installing backdoors for remote access and command-and-control communication with their servers.
4. **Lateral movement:** In this step, attackers move laterally within the network to find valuable assets while avoiding detection. This may involve privilege escalation techniques like Pass-the-Hash attacks, or using stolen credentials to log in as legitimate users.

Data exfiltration and persistence: After locating sensitive data files, documents, credentials, and more, attackers start exfiltrating them from the environment using tunneling techniques or encrypted channels. At the same time, they maintain persistence within the network by deploying backdoors, trojans, and malware.

Examples of Advanced Persistent Threats Attacks

Here are some recent examples of Advanced Persistent Threat (APT) attacks:

- **SolarWinds:** The SolarWinds cyberattack was a significant supply chain attack attributed to APT29 (Cozy Bear), a Russian-state-sponsored APT group. The attackers compromised the SolarWinds Orion software platform, used by thousands of organizations for IT infrastructure management. This enabled the threat actors to infiltrate the networks of multiple high-profile targets, including U.S. government agencies and Fortune 500 companies.
- **Hafnium:** Microsoft discovered a Chinese-state-sponsored APT group called Hafnium, which targeted Microsoft Exchange Server vulnerabilities to gain access to email accounts and exfiltrate sensitive data. Hafnium is known to target organizations in various sectors, including defense, healthcare, and higher education.
- **UNC2452 / Nobelium:** An APT group also involved in the SolarWinds attack, continued its cyber-espionage campaign targeting various organizations. In May 2021, Microsoft disclosed that Nobelium had launched a new wave of attacks using the USAID email system to distribute malicious phishing emails.
- **APT41:** A Chinese-state-sponsored APT group which targeted various industries worldwide, including healthcare, telecommunications, and higher education. In 2020, the U.S. Department of Justice (DOJ) charged five Chinese nationals for their involvement in APT41 activities, including unauthorized access to protected computers and stealing sensitive information.

Mitigating Advanced Persistent Threats

While it is impossible to completely prevent APT attacks, here are a few best practices for protecting your organization.

Establishing Effective Security Policies

To minimize the risk of APT attacks, organizations should create robust security policies covering aspects such as access control, password management, and network segmentation. Periodically assessing and updating security policies is crucial for defending against emerging threats.

Promptly Patching Vulnerabilities

Effective vulnerability management is vital in protecting against APTs. Organizations must ensure their software remains updated by quickly applying patches when vulnerabilities are discovered. Additionally, conducting regular vulnerability scans can help detect potential weaknesses before attackers exploit them.

Constant Monitoring and Incident Response Planning

Having a solid incident response plan (IRP) is crucial for rapidly identifying and addressing security breaches. Constant monitoring allows for early detection of suspicious activities within your network, while an IRP details the steps to take upon identifying an attack.

User Awareness Training

Take the following steps to ensure users are vigilant and aware of cyber threats:

- Instruct employees about common social engineering techniques used by threat actors targeting your organization.
- Raise awareness regarding phishing emails and other malicious communications intended to gain unauthorized access or steal sensitive information.
- Integrate cybersecurity training into employee onboarding procedures and ongoing professional development programs.

Collaboration with Security Communities

Organizations should actively engage in security communities to remain informed about the latest threats and best practices for mitigating them. Exchanging information and collaborating with other organizations can help enhance overall cybersecurity posture across various industries.

Advanced Persistent Threat Protection with HackerOne

The HackerOne Attack Resistance Platform goes beyond Advanced Threat Protection with the ability to prevent threats before they can occur. How? By including human security experts to continuously assess the attack surface for the flaws that bad actors target. With HackerOne, organizations are able to outsmart cybercriminals, extend the reach of their security teams and reduce their exposure to threats.

[Learn more about the HackerOne Attack Resistance Platform](#)

Company

Leadership

Careers

Partners

Newsroom

Contact Us

Knowledge

Center

Application Security

Penetration Testing

Cloud Security

Hacking

Cybersecurity

Attacks

DevSecOps

Resources

Blog

Documentation

Leaderboard

Partner Portal

Resources

Contacted
by a
hacker?

